

(17分) 已知 甲盒子装有2个黑球和1个白球, 乙盒子装有2个白球 (所有黑、白球除颜色外完全相同)
 现从甲、乙盒子中各自任取一个球交换放入另一个盒子, 重复 n 次这样的操作后 ($n \in \mathbb{N}^*$),
 甲盒子中黑球个数记为 X_n , 恰有2个黑球的概率为 a_n , 恰有1个黑球的概率为 b_n .

- (1) 求 a_1, b_1 值 (2分) (2) 求证: $\{2a_n + b_n - \frac{6}{5}\}$ 为等比数列 ($n \in \mathbb{N}^*$), 并求出 $\{a_n\}, \{b_n\}$ 通项公式 (12分) (3) 求 X_n 的分布列及其期望 (其中 $n \in \mathbb{N}^*$ 且 $n \geq 2$) (3分)

(1) (用 $\bullet$ 表示黑球, $\circ$ 表示白球)
 甲: 交换一次后, 若甲中恰有2个黑球
 乙: 那么必是甲中白球与乙中白球交换
 由甲中白球只有1个, 概率为 $\frac{1}{3}$, 乙中全为白球, 概率为1
 则 $a_1 = \frac{1}{3} \times 1 = \frac{1}{3}$
 交换一次后, 若甲中恰有1个黑球, 那么必是甲中黑球与乙中白球交换
 由甲中黑球有2个, $P = \frac{2}{3}$, 乙中全为白球, $P = 1$
 则 $b_1 = \frac{2}{3} \times 1 = \frac{2}{3}$

(2) 根据题意, 由全概率公式:
 交换 $n+1$ 次后, 若甲中恰有2个黑球 ($X_{n+1}=2$)
 有两种可能, $P(X_{n+1}=2) = \sum_{i=1}^2 P(X_n=i) \cdot P(X_{n+1}=2|X_n=i)$
 (i) 在 $X_n=1$ 时, 甲: 甲中白球, 乙: 与乙中黑球换
 $P(X_{n+1}=2|X_n=1) = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$
 (ii) 在 $X_n=2$ 时, 甲: 甲中白球, 乙: 与乙中白球换
 $P(X_{n+1}=2|X_n=2) = \frac{1}{3} \times 1 = \frac{1}{3}$
 若甲中恰有1个黑球 ($X_{n+1}=1$)
 有三种可能, $P(X_{n+1}=1) = \sum_{i=0}^2 P(X_n=i) \cdot P(X_{n+1}=1|X_n=i)$
 (i) 在 $X_n=0$ 时, 甲: 甲中白球, 乙: 与乙中黑球换
 $P(X_{n+1}=1|X_n=0) = 1 \times 1 = 1$
 (ii) 在 $X_n=1$ 时, 甲: 甲中黑与乙中黑换, 乙: 或甲中白与乙中白换
 $P(X_{n+1}=1|X_n=1) = \frac{1}{3} \times \frac{1}{2} + \frac{2}{3} \times \frac{1}{2} = \frac{1}{2}$
 (iii) 在 $X_n=2$ 时, 甲: 甲中黑与乙中白球换
 $P(X_{n+1}=1|X_n=2) = \frac{2}{3} \times 1 = \frac{2}{3}$
 则 $b_{n+1} + \frac{1}{2}a_{n+1} - \frac{3}{5} = (-\frac{1}{5})(b_n + \frac{1}{2}a_n - \frac{3}{5})$
 $b_{n+1} + 2a_{n+1} - \frac{6}{5} = (-\frac{1}{5})(b_n + 2a_n - \frac{6}{5})$
 则 $\begin{cases} b_n + \frac{1}{2}a_n - \frac{3}{5} = (-\frac{1}{5})^{n-1} \cdot \frac{12}{5} \\ b_n + 2a_n - \frac{6}{5} = (-\frac{1}{5})^{n-1} \cdot \frac{2}{5} \end{cases}$ 解得 a_n, b_n

解 a_n, b_n 方法
 方法一: ① $a_{n+1} = \frac{1}{3}a_n + \frac{1}{3}b_n$
 ② $b_{n+1} = -\frac{1}{2}b_n - \frac{1}{3}a_n + 1$
 由题, $2a_{n+1} + b_{n+1} - \frac{6}{5} = \frac{1}{6}(2a_n + b_n - \frac{6}{5})$
 得 $2a_n + b_n = \frac{2}{15}(-\frac{1}{6})^{n-1} + \frac{6}{5}$ ③
 又由 $a_{n+1} = \frac{1}{3}a_n + \frac{1}{3}b_n$ ④
 联立 ③、④ 解得 a_n, b_n
 方法二: ① $a_{n+1} = \frac{1}{3}a_n + \frac{1}{3}b_n$
 ② $b_{n+1} = -\frac{1}{2}b_n - \frac{1}{3}a_n + 1$
 设 $b_{n+1} + \lambda a_{n+1} + \mu = p(b_n + \lambda a_n + \mu)$ ③
 将 ①、② 代入 ③ 式, 则
 $\frac{2\lambda-3}{6}b_n + \frac{\lambda-1}{3}a_n + \mu + 1 = pb_n + p\lambda a_n + p\mu$
 则 $\begin{cases} \frac{2\lambda-3}{6} = p \\ \frac{\lambda-1}{3} = p\lambda \\ \mu+1 = p\mu \end{cases}$ 则 $\begin{cases} \lambda = -\frac{3}{4} \\ p = -\frac{1}{5} \\ \mu = -\frac{6}{5} \end{cases}$ 或 $\begin{cases} \lambda = \frac{2}{5} \\ p = \frac{1}{5} \end{cases}$

(1) 由题, $b_1 = \frac{2}{3} \times 1 = \frac{2}{3}$, $a_1 = \frac{1}{3} \times 1 = \frac{1}{3}$ 2分

(2) 由题, $X_n = 0, 1, 2$. 则甲盒中恰无黑球概率为 $1 - a_n - b_n$.

由 $a_{n+1} = P(X_{n+1} = 2) = P(X_n = 1)P(X_{n+1} = 2 | X_n = 1) + P(X_n = 2)P(X_{n+1} = 2 | X_n = 2)$

其中 $P(X_{n+1} = 2 | X_n = 1) = \frac{2}{3} \times \frac{1}{2} = \frac{1}{3}$, $P(X_{n+1} = 2 | X_n = 2) = \frac{1}{3} \times 1 = \frac{1}{3}$

$P(X_n = 1) = b_n$, $P(X_n = 2) = a_n$

则 $a_{n+1} = \frac{1}{3}a_n + \frac{1}{3}b_n$ 5分

由 $b_{n+1} = P(X_{n+1} = 1) = P(X_n = 0)P(X_{n+1} = 1 | X_n = 0) + P(X_n = 1)P(X_{n+1} = 1 | X_n = 1) + P(X_n = 2)P(X_{n+1} = 1 | X_n = 2)$

其中 $P(X_{n+1} = 1 | X_n = 0) = 1 \times 1 = 1$, $P(X_{n+1} = 1 | X_n = 1) = \frac{2}{3} \times \frac{1}{2} + \frac{1}{3} \times \frac{1}{2} = \frac{1}{2}$, $P(X_{n+1} = 1 | X_n = 2) = \frac{2}{3} \times 1 = \frac{2}{3}$

$P(X_n = 0) = 1 - a_n - b_n$

则 $b_{n+1} = -\frac{1}{2}b_n - \frac{1}{3}a_n + 1$ 8分

由 $2a_{n+1} + b_{n+1} - \frac{6}{5} = 2(\frac{1}{3}a_n + \frac{1}{3}b_n) + (-\frac{1}{2}b_n - \frac{1}{3}a_n + 1) - \frac{6}{5} = \frac{1}{6}(2a_n + b_n - \frac{6}{5})$

又由 $2a_1 + b_1 - \frac{6}{5} = 2 \times \frac{1}{3} + \frac{2}{3} - \frac{6}{5} = \frac{2}{15} \neq 0$

则 $\{2a_n + b_n - \frac{6}{5}\}$ 是首项为 $\frac{2}{15}$, 公比为 $\frac{1}{6}$ 的等比数列 10分

则 $2a_n + b_n - \frac{6}{5} = (\frac{1}{6})^{n-1} (2a_1 + b_1 - \frac{6}{5})$ 则 $2a_n + b_n = \frac{2}{15}(\frac{1}{6})^{n-1} + \frac{6}{5}$

由 $a_{n+1} = \frac{1}{3}a_n + \frac{1}{3}b_n = \frac{1}{3}a_n + \frac{1}{3}(\frac{2}{15}(\frac{1}{6})^{n-1} + \frac{6}{5} - 2a_n) = -\frac{1}{3}a_n + \frac{4}{15}(\frac{1}{6})^{n-1} + \frac{2}{5}$

则 $a_{n+1} - \frac{8}{15}(\frac{1}{6})^{n+1} - \frac{3}{10} = (-\frac{1}{3})(a_n - \frac{8}{15}(\frac{1}{6})^n - \frac{3}{10})$ 又由 $a_1 - \frac{8}{15}(\frac{1}{6})^1 - \frac{3}{10} = -\frac{1}{18} \neq 0$

则 $a_n = \frac{8}{15}(\frac{1}{6})^n - \frac{1}{2}(-\frac{1}{3})^{n+1} + \frac{3}{10}$; $b_n = -\frac{4}{15}(\frac{1}{6})^n + (-\frac{1}{3})^{n+1} + \frac{3}{5}$ 14分

(3) 由 $X_n = 0, 1, 2$, 则 X_n 分布列如下:

X_n	0	1	2
P	$-\frac{4}{15}(\frac{1}{6})^n - \frac{1}{2}(-\frac{1}{3})^{n+1} + \frac{1}{10}$	$-\frac{4}{15}(\frac{1}{6})^n + (-\frac{1}{3})^{n+1} + \frac{3}{5}$	$\frac{8}{15}(\frac{1}{6})^n - \frac{1}{2}(-\frac{1}{3})^{n+1} + \frac{3}{10}$

由 $E(X_n) = 0 \times (1 - a_n - b_n) + 1 \times b_n + 2 \times a_n = 2a_n + b_n$ 16分

又由(2), 则 $E(X_n) = 2a_n + b_n = \frac{2}{15}(\frac{1}{6})^{n-1} + \frac{6}{5}$ ($n \geq 2$ 且 $n \in \mathbb{N}^*$)

评分细则: ① a_n, b_n 各1分.

② a_{n+1} 对应全概率公式1分, 条件概率共1分, 结论1分 共3分 b_n 同理

③ 判断 $\{b_n + 2a_n - \frac{6}{5}\}$ 是等比, 验证首项不为0为1分, 结论1分

④ a_n, b_n 通项公式各2分

⑤ X_n 分布列 2分

⑥ $E(X_n)$ 结果 1分 17分